

Lab 2 - Wireshark Network Traffic Analysis

DNS, TCP Three-Way Handshake, and TLS 1.3 Investigation

Attribute	Details
Tool	Wireshark
Capture interface	Ethernet
Protocols	DNS, UDP, TCP, TLS 1.3, IPv4, IPv6
Local IPv4 observed	192.168.1.6
Exercise type	Authorized network capture

1. Objective

Capture and analyze real network traffic using Wireshark, identify DNS resolution, observe a TCP three-way handshake, and examine TLS 1.3 connection establishment from a SOC analyst perspective.

2. Investigation Procedure

- Captured traffic on the Ethernet interface.
- Used display filters for DNS, TCP, and TLS.
- Followed packet sequences to distinguish name resolution, transport establishment, and encrypted application traffic.

3. DNS Investigation

A DNS query for chatgpt.com originated from 192.168.1.6 and was sent to 192.168.1.1 over UDP port 53. The query used DNS record type HTTPS (65). The response included two AAAA records: 2a06:98c1:3101::6812:202f and 2606:4700:4408::ac40:9bd1.

4. TCP Three-Way Handshake

Step	Packet	Purpose
1	SYN	Client requests a TCP connection
2	SYN, ACK	Server acknowledges and responds
3	ACK	Client confirms; connection established

The handshake was observed for an IPv6 connection to destination port 443, confirming successful TCP establishment before secure application communication.

5. TLS 1.3 Investigation

A TLS 1.3 ClientHello was observed after the TCP handshake. The Server Name Indication identified editor.svc.cloud.microsoft. A later response contained ServerHello and Application Data, showing that TLS negotiation progressed and application content became encrypted.

6. Filters Used

Purpose	Filter
DNS	dns
TCP	tcp
TLS	tls
Connection example	tcp.port == 57462

7. Analyst Assessment

The observed traffic represented a normal sequence of DNS resolution, TCP connection establishment, TLS negotiation, and encrypted application data. No malicious activity was concluded from the capture.

8. Skills Demonstrated

- Packet capture analysis
- DNS investigation

- TCP/IP fundamentals
- TLS 1.3 inspection
- IPv4/IPv6 traffic analysis
- Wireshark filtering

9. Conclusion

The investigation demonstrated how a SOC analyst can use Wireshark to identify protocol stages and distinguish connection setup from encrypted application traffic. The original .pcapng capture should be retained as supporting evidence.